

Debarshee Chakraborty

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I am a Backend and Systems Engineer specializing in Golang, Python, and Java with experience in distributed architectures and backend scalability. My work focuses on problem-solving and managing cloud application services.

EDUCATION

Academy of Technology, B.Tech with Hons. in Computer and Electrical Engineering Oct 2022 – June 2026

- **Relevant Coursework:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design, Networking, Operating Systems, and Machine Learning.

SKILLS

Core Backend & Systems: Go, Python (FastAPI, Django), Distributed Systems, System Design & Scalability, gRPC, RESTful APIs, Java (Spring Boot), Node.js.

AI, ML & Data Pipelines: Agentic AI (LangGraph, LlamaIndex), RAG Pipelines, Vector Embeddings, LLMs (Gemma, Qwen), PyTorch, Hugging Face, CUDA.

Cloud & Infrastructure: AWS, Docker, Kubernetes, CI/CD, TypeScript, React.

Engineering Practices: Performance Tuning, Code Security, Agile Prototyping, Data Structures & Algorithms.

EXPERIENCE

Backend Developer, Resourcio - [Apertre Website](#) Jun 2023 – May 2025

- Established best practices for scalable architecture and led code reviews to ensure high-quality delivery, maintaining 99.9% system availability for 5,000+ active users.
- Reduced leaderboard processing runtime by 87% (from 4 minutes to 30 seconds) as validated via execution logs by refactoring algorithms for multi-node consistency and implementing asynchronous data handling.

PROJECTS

Olake - Datazip (Open Source) [GitHub Repository](#)

- Reduced data replication failures by 95% as quantified through error-rate analytics by engineering resilient retry logic with exponential backoff for MongoDB chunking and splitVector operations.
- Streamlined large-scale data lake ingestion as demonstrated by fault-tolerant replication success rates by implementing database replication logic using Apache Iceberg and Amazon S3.
- Tech Stack: Go, Java, Shell, SQL, Amazon S3, Iceberg

Agentic RAG Pipeline (Applied AI) [GitHub Repository](#)

- Architected a full-stack RAG pipeline using PyTorch and Gemma-7B, integrating LangGraph and LlamaIndex for CUDA-accelerated local retrieval and self-recovering document processing.
- Tech Stack: Python, PyTorch, CUDA, Gemma-7B, LlamaIndex, LangGraph, Vector Databases.

Traffic and Vehicle Management using Object Detection [GitHub Repository](#)

- Accomplished a 60% reduction in traffic signal idle time as measured by real-time predictive analytics and statistical modeling, by doing the development of a system that integrated machine learning with IoT-enabled traffic signal cameras.
- Tech Stack: Python, FastAPI, PyTorch, Hugging Face, Docker, Postgres, SQL, Redis

ACHIEVEMENTS

AWS Cloud Solution Architect - Professional Certificate Mar 2025 – Present

- Learned advanced cloud resource and tool management solutions, how to scale services with minimum overhead, how to handle big data in different scenarios, and advanced cloud tooling.